

EZHILKUMARAN M

+91 8122335610 • ezhilkumaranmanikandans@gmail.com

[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

AI & Data Science undergraduate with hands-on experience building AI agents, machine learning applications, full-stack web platforms, and data analysis solutions. Proficient in **Python, LangChain, PyTorch, RAG pipelines, Flask, React.js**, and **PostgreSQL**. Skilled in REST API development, data analytics, experiment design, and cloud deployment. Passionate about solving real-world problems through scalable software and artificial intelligence.

EDUCATION

Gnanamani College of Technology — Namakkal, India

Aug 2023 – May 2027

B.Tech in Artificial Intelligence and Data Science (Pursuing) | CGPA: **9.03**

Coursework: Machine Learning, Deep Learning, Python Programming, Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Database Systems, Web Technologies

The Gurai Higher Secondary School — Salem, India

Passed Out 2023

HSC — Mathematics and Computer Science | 78%

TECHNICAL SKILLS

Languages: Python, C++, C, SQL

AI / ML: LangChain, RAG, Prompt Engineering, AI Agent Development, Model Evaluation, Neural Networks

Data & Analysis: NumPy, Matplotlib, Dataset Construction, Data Cleaning, PostgreSQL, Vector Databases

Web & Frameworks: Flask, React.js, Streamlit, Tailwind CSS, REST APIs, JWT Authentication, Vite

Tools & Platforms: Git, GitHub, Vercel, Postman, Render, Cloudinary, Pinecone, Microsoft Office, Redis.

Core Concepts: Object-Oriented Programming, Fine-tuning, CRUD Operations, Responsive UI/UX, Cross-browser Compatibility

PROJECTS

DocuMind Enterprise — AI-Powered Enterprise Document Intelligence Platform

2025 – 2026

Python, LangChain, RAG, Vector Database, React.js, Flask, PostgreSQL, Vercel | [Live Demo](#)

- Built an enterprise-grade document intelligence platform enabling natural language Q&A on organisational documents (SOPs, manuals, reports) using a **Retrieval-Augmented Generation (RAG)** architecture with semantic search and LLM-powered responses.
- Designed and implemented end-to-end RAG pipelines — document ingestion, chunking, vector embedding, and **LangChain-orchestrated** response generation; optimised retrieval accuracy through prompt engineering and context window tuning.
- Developed a secure full-stack application: React.js frontend, Flask REST API backend, PostgreSQL database — with user authentication, document management, and a conversational AI interface — deployed to production on Vercel.

StatBot Pro — Autonomous AI Data Analyst Agent

2025 – 2026

Python, LangChain, Pandas, NumPy, Matplotlib, Streamlit | [Live Demo](#)

- Designed and shipped an AI-powered agent that accepts CSV/Excel datasets and answers natural language queries — autonomously generating and executing **Python/Pandas code** to surface insights, statistics, and data visualisations.
- Applied **prompt engineering and LangChain** for multi-step agent orchestration; handled edge cases including malformed inputs, ambiguous queries, and runtime execution errors gracefully.

- Delivered as a fully interactive Streamlit dashboard publicly accessible via URL; validated with real-world datasets across multiple diverse domains.

ShelfMart — Full-Stack E-Commerce Platform

2025 – 2026

React.js, Tailwind CSS, Flask, PostgreSQL, Razorpay, Cloudinary | [Live Demo](#)

- Built a complete bookstore e-commerce platform end-to-end — React.js frontend, **Flask REST API** backend, PostgreSQL database, JWT authentication, cart state management, and an admin dashboard — independently deployed to production.
- Integrated **Razorpay** for demo payments and **Cloudinary** for image hosting; wired all UI components to live API responses with skeleton loading states and user-facing error messages.
- Ensured fully responsive layouts across all screen sizes using Tailwind CSS with a consistent design system; applied cross-browser compatibility checks throughout development.

INTERNSHIP EXPERIENCE

Infotac Solution

Mar 2025 – June 2025

Data Associate L1 — Python Development

- Worked on Python-based data processing and automation tasks; applied data handling techniques to clean, transform, and organise datasets for downstream analytics pipelines within a professional team environment.
- Collaborated with senior developers to deliver scripting tasks consistently on schedule, strengthening practical skills in production-grade data engineering and Python scripting.

Viswa Digital Technology

1 Month, 2024

Web Development Intern

- Built a Food Delivery Application end-to-end (HTML/CSS frontend, Python/SQL backend) and shipped within a professional team workflow — first hands-on experience delivering a web product to production.

True Value Professional Consultancy

3 Months, 2024

Executive Trainee — Admin Department

- Managed documentation and data tasks using Excel and MS Word; contributed to cross-team projects and consistently met deadlines across concurrent assignments, demonstrating strong organisational and communication skills.

CERTIFICATIONS & ACHIEVEMENTS

- **Deep Learning using PyTorch + Machine Learning 101** — GUVI / HCL, 2025 | Course Completion Certification
- **Cloud Computing and Distributed Systems** — NPTEL, 2025 | Elite Grade (66%)
- **Introduction to Industry 4.0 and Industrial IoT** — NPTEL, 2026 | Elite + Silver (80%)
- **Freshers Code War v1.0** — Park College of Engineering & Technology, 2025 | Competitive Coding Certification